0x45.dev

BUILD LOG #1 · GUIDE

How I built my AI avatar

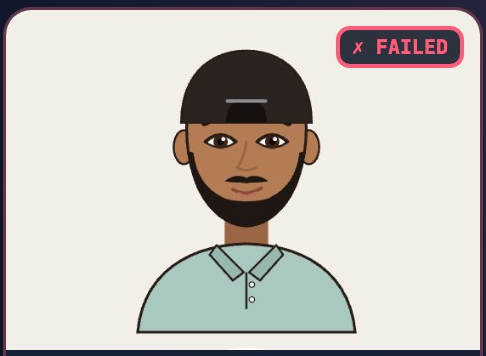
The full flow behind the 0x45.dev presenter: a 3D look of me, my cloned voice, and a face that talks from the audio. **Free tools, running on a laptop.**

4 ATTEMPTS | 3 FAILED | 1 WORKS | $0 TOOLS



WHAT FAILED, WHAT WORKED

4 tries. 3 failed.



ATTEMPT 1

Drew myself in code (SVG)

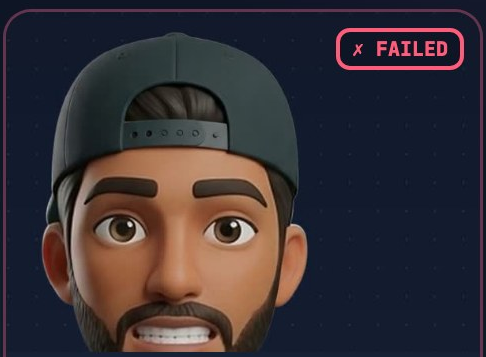
Looked uncanny. Deleted.



ATTEMPT 2

AI 3D look + whole mouth pictures swapped

Beard and lighting flickered.



ATTEMPT 3

One image, only a mouth patch (Rhubarb)

OK for drafts; mouth opens too wide.



ATTEMPT 4

AI animates the whole face from audio

Head, eyes and lips move together.

🎙 voice → JoyVASA · motion → LivePortrait · render → 🙂 face

The one that works is the one you're watching 👆



THE LOOK

A 3D cartoon of you

Gemini app (free) rembg

1. Open the free **Gemini** app and attach **3-4 clear photos** of yourself: front, slightly left, slightly right.

2. Ask for the character:

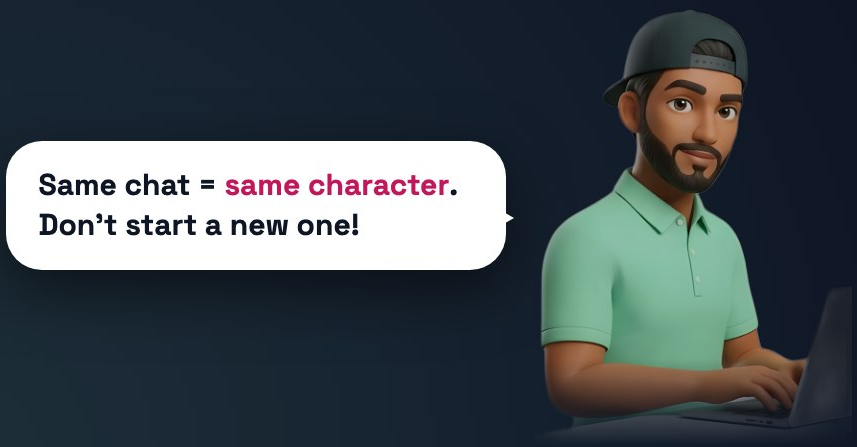
PROMPT

```
Create a Pixar-style 3D character of this person,
chest-up, facing the camera, neutral expression, mouth
closed, plain light-grey background, soft studio light.
Keep the face, beard and hairstyle recognisable.
```

3. In the **same chat**, ask for variations from that image: smile, wink, shocked, thinking, and poses (pointing, thumbs-up). **Same chat = same character.**

4. Remove the background with any free remover (or `rembg`). Keep the original with the grey background too: the face step uses it.

💡 Oversize the eyes slightly and keep a clear lip area. If the beard covers the mouth, lip-sync won't read.



THE VOICE

Clone yourself from a voice note

Chatterbox (MIT) · Whisper · DeepFilterNet

1. Record a **60-90 s voice note**, talking naturally in a quiet room.
2. Cut a clean **10-15 s** piece: no music, no echo.
3. Use **Chatterbox** (Resemble AI) with that clip as the reference voice. Write numbers and odd names the way you say them: `"n p x"`, `"five point five"`.
4. Transcribe every take with **Whisper**. If it hears different words, generate that line again. Big numbers and short words garble most.
5. Remove hiss with **DeepFilterNet**, then normalise the voice to `-16 LUFS`.

> 💡 Slow pace + 0.4 s pauses between lines = every word clear. Nobody complains a Short is too clear.



THE FACE

Make one image talk

JoyVASA (MIT) LivePortrait (MIT)

JoyVASA turns the audio into head, eye and lip motion; **LivePortrait** renders your one image with that motion. Add a **speech gate** so the mouth closes in pauses, remove the grey background with a **colour key**, and export a transparent video.

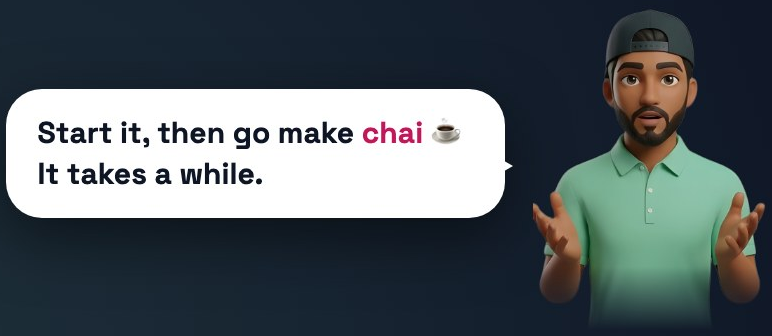
~20	1	$0
min of compute per video	source image	tools & models

PUT IT TOGETHER

Edit it with code

Remotion (React) FFmpeg

Use **Remotion** to place the talking head, word-timed captions (from Whisper), cards and music, then export `1080×1920` for Reels/Shorts. Master the final audio to `-14 LUFS`.



EVERYTHING I USED

13 tools, $0

✨ **Gemini app** free tier
The 3D cartoon look from my photos
gemini.google.com

✂️ **rembg** MIT
Background removal for the cut-outs
github.com/danielgatis/rembg

🔍 **Real-ESRGAN** BSD
4× upscale so the face is sharp in HD
github.com/xinntao/Real-ESRGAN

🎙️ **Chatterbox** MIT
Voice clone from a short voice note
github.com/resemble-ai/chatterbox

👂 **Whisper** MIT
Captions + checks every voice take
github.com/openai/whisper

🧹 **DeepFilterNet** MIT/Apache
Removes hiss from the voice
github.com/Rikorose/DeepFilterNet

🧠 **JoyVASA** MIT
Audio → head, eye and lip motion
github.com/jdh-algo/JoyVASA

🙂 **LivePortrait** MIT
Renders the one image talking
github.com/KlingAIResearch/LivePortrait

👄 **Rhubarb Lip Sync** MIT
Mouth shapes (draft avatar)
github.com/DanielSWolf/rhubarb-lip-sync

🎹 **FluidSynth** LGPL
Plays the code-written lo-fi music
github.com/FluidSynth/fluidsynth

⚛️ **Remotion** free for individuals
Edits the video as React code
remotion.dev

🎬 **FFmpeg** LGPL/GPL
Encoding + loudness mastering
ffmpeg.org

🤖 **Claude Code** Anthropic
Wrote and runs the whole pipeline with me
claude.com/claude-code

Most of these are MIT licensed. Free, even for your channel 👍



PLAY FAIR

Licences to respect

⚠ Only clone **your own voice**, or voices you have permission for.

⚠ Some face-detection models (e.g. **InsightFace**) are **non-commercial**: avoid them for monetised videos.

⚠ Label the video as **AI-generated** where the platform asks (YouTube: "altered or synthetic content").

0x45.dev

✦ AI

AI updates, security fixes and a trending repo every day, **explained fast**.

Follow @0x45.dev

Made by Nouman, a real developer, and his AI avatar.



Questions? Comment on any video. See you there!